

HOW AN ABC TRANSPORTER – TWO-COMPONENT SYSTEM MEDIATES ANTIMICROBIAL PEPTIDE RESISTANCE IN *STREPTOCOCCUS* *PNEUMONIAE* ?

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ABSTRACT

Antimicrobial peptides (AMPs), produced by most living organisms, are promising antibacterial compounds and constitute a key component of innate immunity. Most AMPs act by disrupting the bacterial membrane or by inhibiting peptidoglycan biosynthesis, an essential process for bacterial cell wall integrity (1). In *S. pneumoniae*, a pathogen responsible for severe respiratory infections, resistance to AMPs is mediated by the BceAB/TCS01 signaling system, which couples an ABC transporter to a two-component regulatory system. AMP sensing by BceAB activates a phosphorylation cascade, resulting in the induction of *bceAB* expression and enhanced resistance to peptidoglycan-targeting AMPs (2,3). Also, TCS01 has been identified as a major virulence determinant in *S. pneumoniae* (4). Despite its importance, the molecular mechanism underlying BceAB/TCS01 function remains poorly understood. This project aims to elucidate the molecular basis of BceAB/TCS01-mediated AMP resistance and improve our understanding of host–pathogen interactions.

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